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2 DOCTORAL THESIS

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4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

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7 *Author:*

Tahir JAVAID

Supervisor:

Prof. Guoming CHEN

Co-supervisor:

Prof. Mingshui CHEN

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Abstract

Institute of High Energy Physics
Chinese Academy of Sciences

Doctor of Science

The study of Higgs properties in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \rightarrow 4\ell$ decays ($\ell = e, \mu$) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV. The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to integrated luminosity of 137.1 fb^{-1} at 13 TeV (Full Run 2 data). These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73_{-0.29}^{+0.30} = 2.73_{-0.22}^{+0.23}(\text{stat.})_{-0.19}^{+0.24}(\text{syst.}) \text{ fb}$ at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

All the integrated and differential measurements are compared with the Standard Model based theoretical predictions and found to be consistent within their uncertainties.

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Chapter 1

Introduction

After a quest, ordinary matter and underlying forces or interactions (electromagnetic, weak and strong interactions and not the gravity) which bind it in a structure are successfully explained by the Standard Model (SM) of the particle physics. In SM, description of these interactions is made by symmetry laws of mathematics which could only explain the characteristic of particles with zero mass. Higgs mechanism was then introduced to make the theory realistic enough to incorporate mass problem. Under this mechanism, electroweak symmetry breaks and as a result particles gain the mass. Higgs boson is the direct manifestation of this Higgs mechanism [1, 2, 3, 4, 5, 6].

Precision measurements on the high energy physics experiments like Tevatron and LEP have proven the authenticity of SM; however, Higgs observation could not happen. In 2010, Large Hadron Collider (LHC), the largest experiment ever of its kind in physics world, was then set to put more tests on SM and it endorsed the measurements on SM from earlier experiments. In 2012, the “hidden” Higgs boson, was discovered in shape of heavy scalar boson as jointly reported by the CMS and ATLAS experiments [7, 8]. With this happening, physicists were all set to measure other properties of Higgs for further tests on SM predictions. Mass of Higgs boson was only free parameter in Higgs sector of SM until measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [9].

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of SM at the TeV energy scale. Any deviation of the

integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. These measurements with earlier or partial Run2 CMS data has also been performed and reported in Ref. [Khachatryan:2015yvw, CMS:2015hja, CMS:2016rqf, Sirunyan:2017exp, CMS:2018hhg]. Nearly similar measurements have already been reported by ATLAS experiment in $H \rightarrow \gamma\gamma$ [atlas-conf-2019-029, 10, 11] and $H \rightarrow 4\ell$ [atlas-conf-2019-025, 12, 13] decay channels, and combination of both channels [atlas-conf-2019-032, 14, 15], and by CMS experiment in $H \rightarrow \gamma\gamma$ decay channel [CMS:2019drq, 16]. More precise Higgs boson properties measurements are needed to spot any possible deviation from SM predictions which can be a doorway to new physics.

The dissertation is built on inclusive and differential fiducial cross sections which are measured using the full Run 2 CMS data recorded at 13 TeV corresponding to several kinematic observables which are sensitive to the H boson production mechanisms.

This thesis is structured as followed:

- **Chapter 1 ??:** An overview of theoretical background i.e. the SM of particle physics, production and decays of H boson are described.
- **Chapter 2 ??:** Overview of relevant experiment i.e. the LHC and its CMS experiment are presented.
- **Chapter 3 ??:** The calibrations and reconstructions of the physics objects within the CMS detector are described.
- **Chapter 4 ??:** Presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV.
- **Chapter 5 ??:** Finally, the conclusions and outlook of the dissertation are summarized.

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